SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	S.Harish	**Reg. No.:**	192524413
Section / Batch:	B.Tech(AIDS)	**Date:**	17.7.2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
		GRAND TOTAL:	**100 Marks**

SECTION A – APPLICATION BASED QUESTIONS

Q1. Application: Weather Forecasting Simulation System

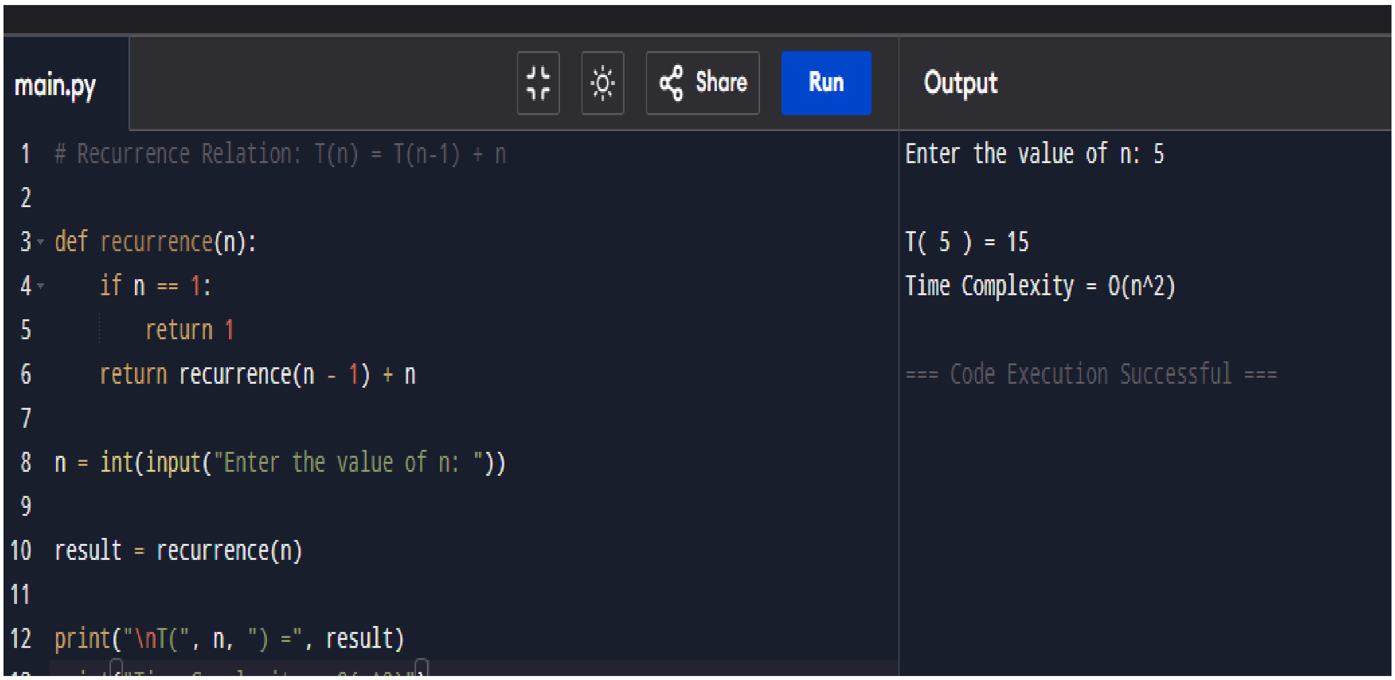
A weather simulation system uses two algorithms with complexities O(n log n) and O(n^2 log n). Analyze both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

```python
import math
n = int(input("Enter the value of n: "))

# Time complexity calculations
algorithm_A = n * math.log2(n)
algorithm_B = (n ** 2) * math.log2(n)

print("\nAlgorithm A (O(n log n)) =", round(algorithm_A, 2))
print("Algorithm B (O(n^2 log n)) =", round(algorithm_B, 2))

# Compare algorithms
if algorithm_A < algorithm_B:
    print("\nAlgorithm A is more efficient for large datasets.")
else:
    print("\nAlgorithm B is more efficient.")
```

```
Enter the value of n: 5

Algorithm A (O(n log n)) = 11.61
Algorithm B (O(n^2 log n)) = 58.05

Algorithm A is more efficient for large datasets.

=== Code Execution Successful ===
```

Q2. Application: Online Learning Recommendation System

An online learning platform processes user interaction data using the recurrence T(n) = T(n−1) + n. Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using asymptotic notation and discuss whether the system is scalable for large educational platforms.



Code of Conduct

I _______S.Harish(192524413)_________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________S.Harish______________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem

Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________